An RF-based Low Rate DDoS Attack Real-time Detection System

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***Abstract*—Software-Defined Networking (SDN) is a new network architecture paradigm. It improves flexibility, scalability and management of the network by separating forwarding logic and control logic. SDN controllers offer the ability to dynamically modify traffic forwarding rules and have a comprehensive view of the entire network. However, Introducing SDN brings some new vulnerabilities to DDoS attacks, such as single point of failure. Attacks known as Distributed Denial of Service (DDoS) overload network links with illegal traffic at high speeds. Illegal traffic consumes network bandwidth, resulting in legal packets lost and network services unavailable. Low Rate Distributed Denial of Service (LRDDoS) attack is a variant of DDoS attacks and is becoming one of the most serious vulnerabilities. LRDDoS attacks consume network resources by periodically sending a relatively small number of packets. Consequently, it is harder to detect LRDDoS attacks.**

This paper proposes a real-time system for LRDDoS defense in SDN. The system uses Random Forest(RF) to detect attacks. We call this system as RFLRS. In addition, based on the CIC dataset, we suggest a feature subset that is most suitable for RFLRS to speed up classification. The experimental results show that RF performs a 99.8% accuracy with a reduction of 28% classification time.

***Index Terms*—Software-Defined Networking, Low Rate Distributed Denial of Service, Random Forest, Real-time System,**

I. Introduction

The network has become an indispensable part of modern life for sharing information and resources using digital information technology. It is a process of end-to-end communication by using forwarding devices. However, the architecture of traditional networks is relatively static and inflexible, making it difficult to reformulate and reconfigure new protocols and rules in the network. An important reason is that the data plane and the control plane are strongly coupled, and the protocols for control and routing are embedded in the physical data forwarding devices. The high degree coupling between the data plane and the control plane makes it more difficult to dynamically manage the network and its protocols. Whenever any existing protocol needs to be updated or new functions need to be added, the configuration of all affected devices must be modified [1].

SDN is a novel network architecture that has been proposed to improve the shortcomings of the existing network infrastructure [2]. Network switches become simple forwarding devices as a result of the separation of the control and data planes, and control logic is implemented in an effectively centralized controller. The controller centrally controls and manages the network functions and instructs forwarding behavior to all elements distributed over the infrastructure. This characteristic of SDN makes it different from traditional network technologies, which tightly combines the control plane and data plane. Using this idea makes it possible to manage the network state by software programming [3]. In exchange for these advantages, the controller-centric SDN architecture is more vulnerable to the risk of single points of failure caused by traffic overload than traditional networks [4].

DDoS attacks are considered to be the most destructive attacks [5]. DDoS attacks are capable of causing significant damage to Information and Communication Technology(ICT) infrastructures. DDoS attacks aim to consume the network bandwidth by making a large number of illegal packets. SDN-targeted DDoS attacks inflict more severe destruction to network infrastructure compared to traditional networks. In SDN, switches use messages to notify the controller to obtain new traffic forwarding instructions for any data flows that they do not know how to handle. Because of this mechanism, a DDoS attack may generate lots of new packets, then switches will generate many messages to send to the controller. As a result, the controller's resources, including its CPU and memory, will be exhausted, and the controller will fail [6].

DDoS attacks have progressed in recent years from simple traffic to fast traffic to comparatively slow traffic [7]. As DDoS attacks have evolved, a new security challenge called Low Rate DDoS (LRDDoS) has appeared [8]. Different from traditional DDoS attacks which overwhelm the target with traffic, LRDDoS attackers generate attack traffic with relatively low speeds. Instead of completely overloading the

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target, LRDDoS attacks aim to exhaust its resources, such as bandwidth, computing power, or memory. Attack traffic is disguised as legal traffic and hidden in the background.

This paper proposes a system for real-time LRDDoS attack detection and mitigation on SDN that makes use of Random Forest (RF). In the subsequent sections, we call this system as RFLRS. The RFLRS consists of three parts. The first part is the feature extraction model, which collects network traffic information in real-time and selects appropriate features for classifying LRDDoS or normal traffic. The second part is an RF-based packet detection model, which classifies network traffic into normal traffic groups and LRDDoS traffic groups based on the features extracted in the first step. Finally, the mitigation module issues SDN traffic rules to drop packets depending on the classification results for attack traffic. In addition, this paper proposes a feature set applicable to RFLRS to train and test RF models based on importance. This feature set is calculated from the CIC dataset [9].

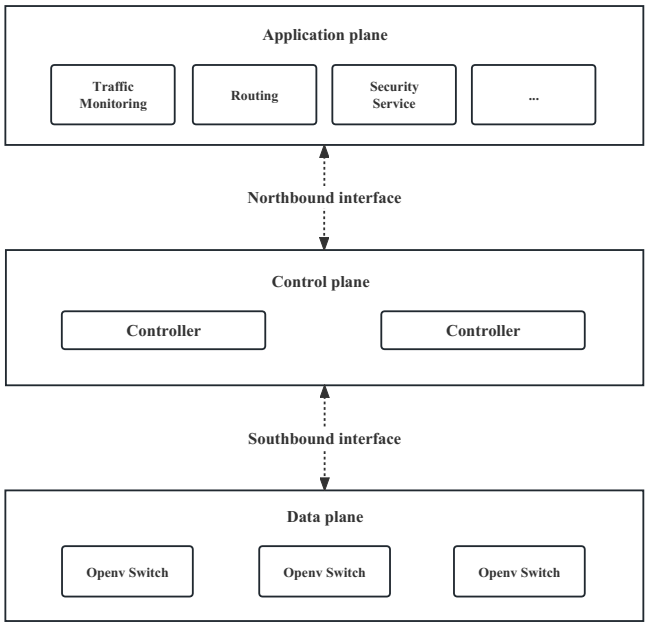

Fig. 1. SDN architectural view

II. Background

We discuss the SDN architecture and the threat of LRDDoS attacks in this section.

A. OpenFlow-based SDN architecture

An SDN architecture is shown in Figure 1, which is divided into three layers: data plane, control plane and application plane.

The data plane contains many switches. An SDN switch is only responsible for forwarding packets according to the instructions from the controller. The key to achieving this function is the OpenFlow protocol [10]. According to the definition of OpenFlow, each switch is required to maintain one or more flow tables. Each flow table entry contains part of the packet's information for making forwarding decisions. Rule, action and counter are the three key parts of every entry in the flow table. The rule determines the packet header fields' values. The switch searches the flow table for rules whenever it receives a new data flow. If the field value matches, the counter value is raised and the switch will perform the necessary action (for example, forward the packet or drop it). Similarly, if the field values do not match, the switch will send a *packet_in* message to the corresponding SDN controller to notify it. By sending OpenFlow's *flow_mod* message, the controller decides which action is suitable and adds a new rule to the flow table in switches.

One or more controllers make up the control plane. It is in charge of managing the entire network as well as controlling all switches. OpenFlow is the most significant of these standard southbound interfaces for communication between the SDN controller and switches.

The application plane contains several different applications with business concerns and needs. SDN applications can communicate with the controller using the northbound interface, which is typically in the form of RESTFull API [11]. Applications can call the northbound interface to obtain information about the data plane and directly interfere with the packet-forwarding process by generating and issuing flow rules.

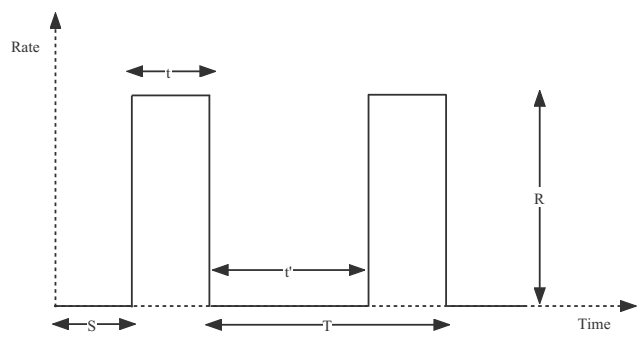

Fig. 2. LRDDoS attack stream

B. Periodic Low Rate DDoS Attack

The LRDDoS attack is one intelligent attack that has the characteristics of a periodicity [12]. This LRDDoS attack is typically illustrated by a periodic waveform shown in Figure 2. Five parameters can broadly be used for describing an LRDDoS attack: Attack period (T), burst width (t), attack idle time (t'), number of packets sent in a single attack pulse (R) and attack start time (S) [13].

The successful LRDDoS attacks usually have the characteristic that T is the minimum Retransmission TimeOut(RTO) value or the integer times of the RTO value [14]. [15] proposed an LRDDoS attack known as the Slow TCAM Exhaustion attack for the SDN data plane. This is an LRDDoS attack targeting the SDN data plane to consume the capacity of the

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switch's flow table slowly. [16] proposed a new Slow TCAM Exhaustion attack with linearly increasing attack traffic. This attack will increase the number of attack flows in each attack period, which causes the switch to install more flow table entries until the flow table is saturated. Therefore, it is crucial to mitigate the attack in as little T as possible. If a LRD-DoS attack cannot be quickly and accurately mitigated, the influence will carry over to the next period. To achieve these goals, faster and more accurate packet inspection methods are essential.

III. Methodology

Artificial intelligence techniques have been extensively utilized in the field of cybersecurity in recent years. Machine learning algorithms can learn from data how to distinguish between normal and attack traffic, and this ability has proven to be very effective in developing a reliable Intrusion detection system (IDS) [17]. This paper uses Random Forest as a means of detecting network traffic and also selects appropriate features to reduce the detection time to mitigate LRDDoS attacks as soon as possible. In addition, the dataset used for evaluation is also introduced in this section.

A. Random Forest

Different machine learning algorithms are applied to the Intrusion Detection (ID) problem and RF is considered one of the most commonly used algorithms [18]. Numerous studies have analyzed the effectiveness of different machine learning algorithms on various benchmark datasets and have discovered that RF has the highest accuracy [19].

RF is an approach for ensemble learning that combines many decision trees to generate a prediction [20]. It builds multiple decision trees for classification and regression tasks and determines the final prediction by majority voting or averaging the results. Each decision tree is independently constructed, and RF introduces two sources of randomness during the tree construction: data sampling and feature selection. After the forest is built, when a new input sample enters, the decision trees in the forest make judgments and determine the sample category based on the decision [21].

Apart from this, we seek a better optimized detection time to meet the requirement of threat mitigation with fewer attack periods T. Feature selection techniques provide a means of reducing the computation time [22]. By introducing this approach, it is possible to accelerate RF execution and lose only a small amount of accuracy by using only part of the features. Our experimental results prove that the detection time can be significantly reduced while maintaining high accuracy by selecting only a few important features. The details of the feature selection phase will be introduced in Section V.

B. Dataset

Many datasets are used in the DDoS attack defense evaluation. The widely-used KDD 99 and NSL-KDD datasets mostly concentrate on conventional DDoS attacks, whereas the CIC DoS 2019 dataset additionally includes reflection and exploitation DDoS attacks. A sizable number of LRDDoS attacks are only captured in the CIC DoS 2017 dataset. [23].

This paper mainly focuses on the CIC 2017 and CIC 2019 datasets. To make the results more generalizable, we use a mixture of both datasets. Thus the training set and the test set contain both high rate and low rate DDoS attack traffic (with the low rate dominating) and have the same 78 features. LRDDoS attack traffic was collected on Wednesday in the CIC 2017 dataset. 692,698 pieces of attack data were recorded over a non-consecutive 74-minute span. The average attack rate is approximately 156 packets per second.

IV. RFLRS design

The RFLRS is designed to provide a capability to monitor network traffic and mitigate LRDDoS attacks in real-time. Figure 3 illustrates the entire architecture of RFLRS.

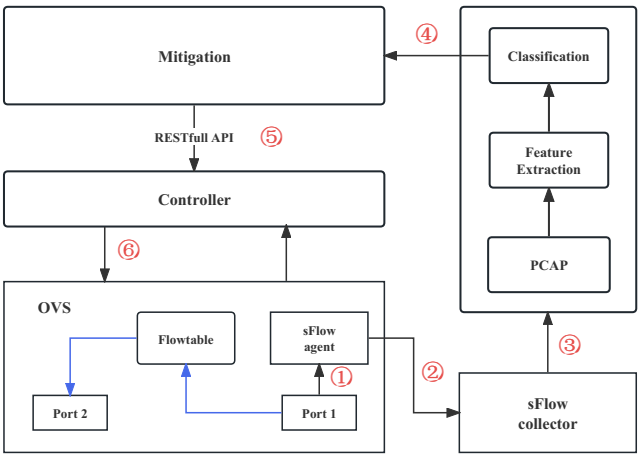

Fig. 3. Architecture of RFLRS

The SDN environment is simulated by the Mininet [24] and Ryu [25] controller. When a data flow enters the OpenFlow switch(OVS) and is forwarded from port 1 to port 2 along the blue line, the steps for mitigating potential threats are as follows:

1) OVS natively supports sFlow monitoring technology, which consists of two components: sFlow agent and sFlow collector. The sFlow agent is deployed in OVS for listening to the ports.
2) The sFlow agent sends passing packet information to the sFlow collector. We select sFlowtool as the sFlow collector and deploy it in the monitor of the detection module.
3) The sFlowtool converts the packet information into PCAP [26] format and stores it. The feature extraction module calculates the features the classifier requires from the PCAP file. An RF-based classifier separates incoming data into normal traffic and attack traffic.
4) The detection module will transmit a signal to the mitigation module containing the attack IP address once it has identified the attack traffic.

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5) The mitigation module obtains the attack IP address and sends an indication to the controller through the northbound interface so that a flow table entry with drop action will be generated.
6) Flow table entries are installed into the switch. In this way, the connection to the attacking host could be disconnected.

V. Experiment and Evaluation

To evaluate the feasibility and performance of RFLRS, we have done some experiments and simulations. This section describes the performance metrics used for the evaluation, the basis for feature selection and the specific simulation environment.

A. Metrics

Common metrics used for evaluating the effectiveness of machine learning algorithms include Accuracy, Precision, Recall, and F1 Score [27]. The degree of machine learning classification approaches can be evaluated using accuracy. Metrics are defined as the following equations (from equation1 to 4). Among these metrics, TN stands for True Negative and refers to instances where the model correctly identified the negative class. Instances where the model correctly identified the positive class are referred to as True Positives (TP). False Negative (FN) cases are those in which the model predicted a negative class, but the actual class was positive. Instances where the model predicted the positive class but the actual class was negative are referred to as False Positives (FP) [28].

The classification time for the test data must be taken into account in addition to these parameters. There are 100,000 rows of data used in the entire experiment, of which 47,945 rows of data are from the CIC 2019 dataset and 52,055 rows are from the CIC 2017 dataset. The training set to test set ratio is split into 6 to 4. The classification time is defined as the time spent to detect 40,000 packets.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP+FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP+FN} \tag{3}$$

$$\text{F1 score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{4}$$

B. Feature Selection

In this paper, feature selection depends on its importance in RF. The concept of ”importance” in RF refers to the importance or contribution of features. It measures which features have the most influence on the prediction results in RF. Importance is determined by evaluating the contribution of each feature to the predictive performance during the tree building process of RF [29]. By computing feature importance, we can understand which features in the dataset have the most influence on the prediction results. Table I shows the top 50 features of importance in the dataset.

TABLE I
Features by importance

Feature	Importance	Feature	Importance
Fwd Packet Length Mean	0.1014	Max Packet Length	0.0127
Fwd Packet Length Min	0.0846	Subflow Fwd Bytes	0.0127
Min Packet Length	0.0834	Flow IAT Mean	0.0125
Average Packet Size	0.0471	Packet Length Variance	0.0125
Avg Fwd Segment Size	0.0456	Flow IAT Max	0.0124
Init_Win_bytes_forward	0.0353	Packet Length Std	0.0117
min_seg_size_forward	0.0351	Bwd Packet Length Max	0.0107
Flow Bytes/s	0.0336	Bwd Packet Length Mean	0.0096
Avg Bwd Segment Size	0.0289	ACK Flag Count	0.0096
Bwd Header Length	0.0284	Flow Packets/s	0.0079
Fwd Packet Length Max	0.0221	Fwd IAT Max	0.0066
Fwd Header Length	0.0220	URG Flag Count	0.0061
Subflow Bwd Packets	0.0219	Fwd IAT Std	0.0057
Packet Length Mean	0.0217	Bwd IAT Total	0.0057
Total Length of Bwd Packets	0.0211	Bwd IAT Std	0.0055
Bwd Packets/s	0.0208	Flow IAT Std	0.0055
Subflow Bwd Bytes	0.0207	Fwd IAT Total	0.0051
Protocol	0.0199	Fwd Packet Length Std	0.0046
Init_Win_bytes_backward	0.0187	Flow IAT Min	0.0042
Fwd Header Length.1	0.0187	Fwd IAT Min	0.0037
Total Length of Fwd Packets	0.0159	Fwd IAT Mean	0.0037
Total Backward Packets	0.0150	Down/Up Ratio	0.0036
Flow Duration	0.0147	Bwd IAT Min	0.0034
Fwd Packets/s	0.0146	Bwd IAT Max	0.0031
Max Packet Length	0.0127	CWE Flag Count	0.0031

C. Result and Discussion

The experimental testbed is built on a host running Windows 11, CPU is Core i7-12700H 2.70 GHz with 16GB RAM. Figure 4 shows the variation of Accuracy, Precision, Recall, and F1 Score when the number of used features increases. ”Top n Features” means in this experiment, the top n features with the highest importance are used to train and test the model. For example, n=2 means that the used features are ”Fwd Packet Length Mean” and ”Fwd Packet Length Min”.

Due to the periodic character of LRDDoS, the defense system needs to detect the attack traffic as fast as possible within the attack pulse time (t) and issue drop rules to mitigate the attack. The classification time is also significant in comparison to the metrics describing the degree of accuracy because it indicates how quickly the RFLRS can respond to an attack. The experimental results are shown in Figure 5.

It can be observed from the combined results of Figures 4 and 5 that when the number of selected features increases, Accuracy, Precision, Recall, and F1 Score will stabilize at a higher level. However, the classification time will fluctuate. The shortest classification time is 0.2901 seconds when using the top 25 high importance features. 28% less than the 0.403 seconds using all 78 features. So the first 25 features in Table 1 are the most suitable feature sets for RF classifier.

D. Simulation

We simulated the RFLRS on an Ubuntu virtual machine. Mininet generated a network topology as shown in Figure 6. Host 4 and Host 6 act as attackers sending low rate attack traffic to Host 1 by using hping3, and the sflow agent is deployed in Switch 1. We did not generate the same traffic as the CIC dataset due to performance reasons. In Figure 7, each row is information of a packet that was captured on switch 1. The first two rows of Figure 8 show the flow table entries that are issued to switch 1 with the action “drop”.

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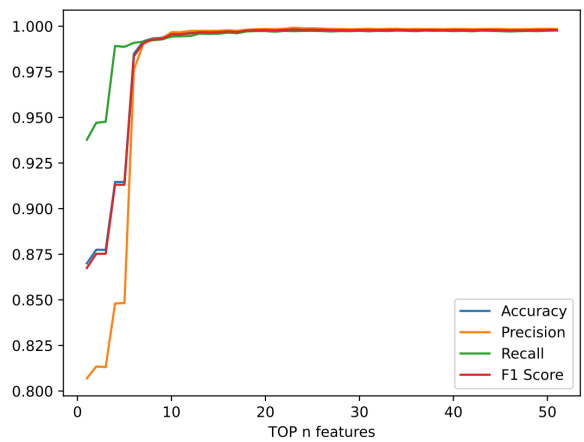


Fig. 4. Performance by TOP Important Features

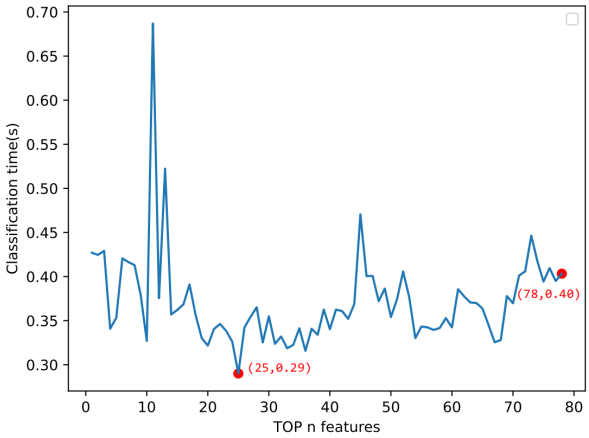


Fig. 5. Classification Time by TOP important features

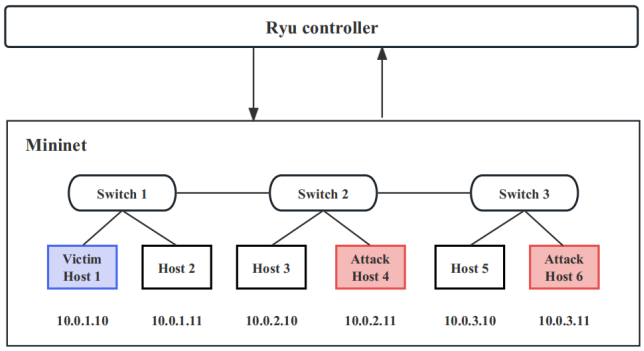
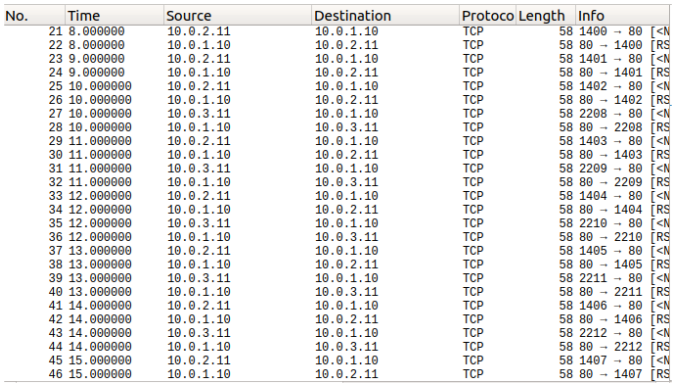


Fig. 6. SDN Topology

VI. Related Work

[30] propose an approach for cloud computing DDoS attack detection. In the proposed work, the authors utilize the Mutual Information (MI) and RF Feature Importance (RFFI) approaches to select the most pertinent features. Select features are tested by RF, Gradient Boosting (GB), Weighted Voting Ensemble (WVE), K Nearest Neighbor (KNN), and Logistic Regression (LR). According to the experimental findings, When using 19 features, RF, GB, WVE and KNN achieved an accuracy of 0.99. However, the detection is only for normal DDoS attacks.



No.	Time	Source	Destination	Protoco	Length	Info
21	8.000000	10.0.2.11	10.0.1.10	TCP	58	1400 → 80 [<N
22	8.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1400 [RS
23	9.000000	10.0.2.11	10.0.1.10	TCP	58	1401 → 80 [<N
24	9.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1401 [RS
25	10.000000	10.0.2.11	10.0.1.10	TCP	58	1402 → 80 [<N
26	10.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1402 [RS
27	10.000000	10.0.3.11	10.0.1.10	TCP	58	2208 → 80 [<N
28	10.000000	10.0.1.10	10.0.3.11	TCP	58	80 → 2208 [RS
29	11.000000	10.0.2.11	10.0.1.10	TCP	58	1403 → 80 [<N
30	11.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1403 [RS
31	11.000000	10.0.3.11	10.0.1.10	TCP	58	2209 → 80 [<N
32	11.000000	10.0.1.10	10.0.3.11	TCP	58	80 → 2209 [RS
33	12.000000	10.0.2.11	10.0.1.10	TCP	58	1404 → 80 [<N
34	12.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1404 [RS
35	12.000000	10.0.3.11	10.0.1.10	TCP	58	2210 → 80 [<N
36	12.000000	10.0.1.10	10.0.3.11	TCP	58	80 → 2210 [RS
37	13.000000	10.0.2.11	10.0.1.10	TCP	58	1405 → 80 [<N
38	13.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1405 [RS
39	13.000000	10.0.3.11	10.0.1.10	TCP	58	2211 → 80 [<N
40	13.000000	10.0.1.10	10.0.3.11	TCP	58	80 → 2211 [RS
41	14.000000	10.0.2.11	10.0.1.10	TCP	58	1406 → 80 [<N
42	14.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1406 [RS
43	14.000000	10.0.3.11	10.0.1.10	TCP	58	2212 → 80 [<N
44	14.000000	10.0.1.10	10.0.3.11	TCP	58	80 → 2212 [RS
45	15.000000	10.0.2.11	10.0.1.10	TCP	58	1407 → 80 [<N
46	15.000000	10.0.1.10	10.0.2.11	TCP	58	80 → 1407 [RS

Fig. 7. Information of Captured Traffic

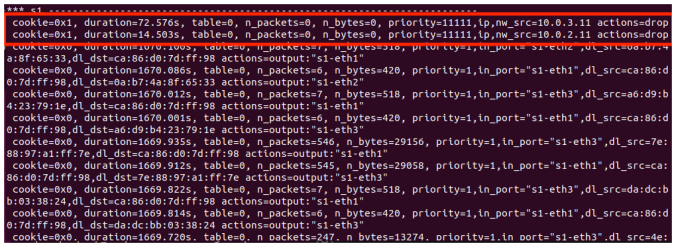


Fig. 8. Flowtable Entries in Switch1

Wu et al. [16] propose a Factorization Machine (FM)-based multi-feature DDoS attack detection approach. Features calculated from flow rules are used to characterize LRD-DoS attacks and an FM machine learning-based approach is implemented for LRDDoS attack detection. However, the way of extracting features from the issued flow rules relies on the controller's reaction to the attacking flow, which has hysteresis. The processing capability of OpenFlow switches is limited. Ternary Content-Addressable Memories (TCAMs) are commonly used to store flow table rules. The cost of using this type of memory is very high [31], so OpenFlow switches only have a limited amount of TCAM, typically storing only 1500-3000 flow rules [32].

[13] propose a modular, adaptable architecture that enables the detection and mitigation of LRDDoS threats in SDN environments. The J48, Random Tree, REP Tree, RF, Multi-Layer Perceptron, and Support Vector Machines machine learning models have been utilized by the authors to train the intrusion detection system (IDS) and to evaluate the performance of the IDS using the CIC DoS dataset. Despite the difficulty in detecting LR-DoS attacks, the evaluation's results indicate that their approach achieves a 95% detection rate. However, this article does not show specific detection or classification times.

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VII. Conclusion

The paper proposes an architecture for a real-time defensive system that mitigates LRDDoS attacks in SDN. The system is capable of continuously detecting packets in the data plane and mitigating LRDDoS attacks. The packet monitoring and information collection process are not dependent on the controller. RF is deployed in the system as a high accuracy classifier. Additionally, we propose a feature set based on the CIC dataset for the purpose of mitigating the impact of attacks at an early stage. According to our experimental results, it can be concluded that by applying this feature set, RF achieved 99.8% accuracy and reduced classification time by 28%. This indicates that RFLRS has the ability to mitigate LRDDoS attacks effectively.

References

[1] R. Swami, M. Dave, and V. Ranga, "Software-defined networking-based ddos defense mechanisms," *ACM Computing Surveys (CSUR)*, vol. 52, no. 2, pp. 1–36, 2019.

[2] N. McKeown, "How sdn will shape networking," *Open Networking Summit*, 2011.

[3] H. Kim and N. Feamster, "Improving network management with software defined networking," *IEEE Communications Magazine*, vol. 51, no. 2, pp. 114–119, 2013.

[4] N. Z. Bawany, J. A. Shamsi, and K. Salah, "Ddos attack detection and mitigation using sdn: methods, practices, and solutions," *Arabian Journal for Science and Engineering*, vol. 42, no. 2, pp. 425–441, 2017.

[5] C. Douligeris and A. Mitrokotsa, "Ddos attacks and defense mechanisms: classification and state-of-the-art," *Computer networks*, vol. 44, no. 5, pp. 643–666, 2004.

[6] D. Li, C. Yu, Q. Zhou, and J. Yu, "Using svm to detect ddos attack in sdn network," in *IOP Conference Series: Materials Science and Engineering*, vol. 466, no. 1. IOP Publishing, 2018, p. 012003.

[7] G. Lei, L. Ji, R. Ji, Y. Cao, X. Shao, and X. Huang, "Extracting low-rate ddos attack characteristics: the case of multipath tcp-based communication networks," *Wireless Communications and Mobile Computing*, vol. 2021, pp. 1–10, 2021.

[8] A. A. Alashhab, M. S. M. Zahid, M. A. Azim, M. Y. Daha, B. Isyaku, and S. Ali, "A survey of low rate ddos detection techniques based on machine learning in software-defined networks," *Symmetry*, vol. 14, no. 8, p. 1563, 2022.

[9] A. Yulianto, P. Sukarno, and N. A. Suwastika, "Improving adaboost-based intrusion detection system (ids) performance on cic ids 2017 dataset," in *Journal of Physics: Conference Series*, vol. 1192. IOP Publishing, 2019, p. 012018.

[10] N. McKeown, T. Anderson, H. Balakrishnan, G. Parulkar, L. Peterson, J. Rexford, S. Shenker, and J. Turner, "Openflow: enabling innovation in campus networks," *ACM SIGCOMM computer communication review*, vol. 38, no. 2, pp. 69–74, 2008.

[11] C. Banse and S. Rangarajan, "A secure northbound interface for sdn applications," in *2015 IEEE Trustcom/BigDataSE/ISPA*, vol. 1. IEEE, 2015, pp. 834–839.

[12] C. Zhang, Z. Cai, W. Chen, X. Luo, and J. Yin, "Flow level detection and filtering of low-rate ddos," *Computer Networks*, vol. 56, no. 15, pp. 3417–3431, 2012.

[13] J. A. Perez-Diaz, I. A. Valdovinos, K.-K. R. Choo, and D. Zhu, "A flexible sdn-based architecture for identifying and mitigating low-rate ddos attacks using machine learning," *IEEE Access*, vol. 8, pp. 155 859–155 872, 2020.

[14] Z. Wu, L. Zhang, and M. Yue, "Low-rate dos attacks detection based on network multifractal," *IEEE Transactions on Dependable and Secure Computing*, vol. 13, no. 5, pp. 559–567, 2015.

[15] T. A. Pascoal, Y. G. Dantas, I. E. Fonseca, and V. Nigam, "Slow tcam exhaustion ddos attack," in *ICT Systems Security and Privacy Protection: 32nd IFIP TC 11 International Conference, SEC 2017, Rome, Italy, May 29-31, 2017, Proceedings 32*. Springer, 2017, pp. 17–31.

[16] W. Zhijun, X. Qing, W. Jingjie, Y. Meng, and L. Liang, "Low-rate ddos attack detection based on factorization machine in software defined network," *IEEE Access*, vol. 8, pp. 17 404–17 418, 2020.

[17] A. S. Khan, Z. Ahmad, J. Abdullah, and F. Ahmad, "A spectrogram image-based network anomaly detection system using deep convolutional neural network," *IEEE Access*, vol. 9, pp. 87 079–87 093, 2021.

[18] P. A. A. Resende and A. C. Drummond, "A survey of random forest based methods for intrusion detection systems," *ACM Computing Surveys (CSUR)*, vol. 51, no. 3, pp. 1–36, 2018.

[19] T. Markovic, M. Leon, D. Buffoni, and S. Punnekkat, "Random forest based on federated learning for intrusion detection," in *IFIP International Conference on Artificial Intelligence Applications and Innovations*. Springer, 2022, pp. 132–144.

[20] Q. Li, Y. Tian, Q. Wu, Q. Cao, H. Shen, and H. Long, "A cloud-fog-edge closed-loop feedback security risk prediction method," *IEEE Access*, vol. 8, pp. 29 004–29 020, 2020.

[21] Y. Chen, J. Hou, Q. Li, and H. Long, "Ddos attack detection based on random forest," in *2020 IEEE International Conference on Progress in Informatics and Computing (PIC)*, 2020, pp. 328–334.

[22] G. Chandrashekar and F. Sahin, "A survey on feature selection methods," *Computers & Electrical Engineering*, vol. 40, no. 1, pp. 16–28, 2014.

[23] CIC Dos Dataset 2017. [Online]. Available: https://www.unb.ca/cic/datasets/dos-dataset.html

[24] L. Bob. Mininet. [Online]. Available: http://mininet.org/

[25] Ryu. [Online]. Available: https://ryu-sdn.org/

[26] TcpdumpGroup. (2010) tcpdump. [Online]. Available: https://www.tcpdump.org/

[27] S. Liu, H. Fukuda, and P. Leger, "Real-time ddos attack defense system in sdn using lssom," in *2023 26th Conference on Innovation in Clouds, Internet and Networks and Workshops (ICIN)*. IEEE, 2023, pp. 69–73.

[28] J. Pei, Y. Chen, and W. Ji, "A ddos attack detection method based on machine learning," in *Journal of Physics: Conference Series*, vol. 1237, no. 3. IOP Publishing, 2019, p. 032040.

[29] C. Strobl, A.-L. Boulesteix, T. Kneib, T. Augustin, and A. Zeileis, "Conditional variable importance for random forests," *BMC bioinformatics*, vol. 9, pp. 1–11, 2008.

[30] M. Alduailij, Q. W. Khan, M. Tahir, M. Sardaraz, M. Alduailij, and F. Malik, "Machine-learning-based ddos attack detection using mutual information and random forest feature importance method," *Symmetry*, vol. 14, no. 6, p. 1095, 2022.

[31] K. Kannan and S. Banerjee, "Compact tcam: Flow entry compaction in tcam for power aware sdn," in *International conference on distributed computing and networking*. Springer, 2013, pp. 439–444.

[32] R. Kandoi and M. Antikainen, "Denial-of-service attacks in openflow sdn networks," in *2015 IFIP/IEEE International Symposium on Integrated Network Management (IM)*. IEEE, 2015, pp. 1322–1326.

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